Tianzhe Chu

Email : tianzhechu@gmail.com
Web : https://tianzhechu.com
Google Scholar: Tianzhe Chu

Education

- **The University of Hong Kong** — Hong Kong SAR
  Ph.D. student at IDS advised by Prof. Yi Ma — *Sep 2024 - Now*
  Research: Foundation Models, Deep Representation Learning, Reinforcement Learning.
- **ShanghaiTech University** — Shanghai, China
  B.Eng in Computer Science and Technology (Honor) — *Sep 2020 - Jun 2024*
  Selected Courses: Introduction to Machine Learning, Probability and Statistics, Computer Architecture I, Data Structure and Algorithms, Discrete Mathematics, Signals and Systems, Mathematical Analysis.
- **University of California, Berkeley** — Berkeley, CA, US
  Visiting undergraduate in EECS — *Aug 2022 - May 2023*
  Graduate Level Courses: Deep Learning, Deep Reinforcement Learning, Foundation of Graphics, Applications of Parallel Computing, Computer Vision, Optimization Model in Engineering

Research Interest

- Foundation models: data, architecture, training, and understanding.
- Any interesting applications.

Publications

(* means equal contribution)

- **Tianzhe Chu***, Yuexiang Zhai*, Jihan Yang, Shengbang Tong, Saining Xie, Dale Schuurmans, Quoc V.Le, Sergey Levine, Yi Ma, *SFT Memorizes, RL generalizes: A Comprehensive Study of Foundation Model Post-training*, https://arxiv.org/abs/2501.17161v1,
  Under Review
- Yaodong Yu*, **Tianzhe Chu***, Shengbang Tong, Ziyang Wu, Druv Pai, Sam Buchanan, Yi Ma, *Emergence of Segmentation with Minimalistic White-Box Transformers*, https://arxiv.org/abs/2308.16271.
  Conference on Parsimony and Learning (CPAL) 2024 (Oral), https://cpal.cc,
  NeurIPS 2023 XAI Workshop (Oral), (4 orals of 59 posters)
- **Tianzhe Chu***, Shengbang Tong*, Tianjiao Ding*, Xili Dai, Benjamin D. Haeffele, René Vidal, Yi Ma, *Image Clustering via the Principle of Rate Reduction in the Age of Pretrained Models*, https://arxiv.org/abs/2306.05272,
  ICLR 2024
- Yaodong Yu, Sam Buchanan, Druv Pai, **Tianzhe Chu**, Ziyang Wu, Shengbang Tong, Benjamin D. Haeffele, Yi Ma, *White-Box Transformers via Sparse Rate Reduction*, https://arxiv.org/abs/2306.01129.
  NeurIPS 2023
- Yaodong Yu, Sam Buchanan, Druv Pai, **Tianzhe Chu**, Ziyang Wu, Shengbang Tong, Hao Bai, Yuexiang Zhai, Benjamin D. Haeffele, Yi Ma, *White-Box Transformers via Sparse Rate Reduction: Compression Is All There Is?*, https://arxiv.org/abs/2311.13110.
  Journal of Machine Learning Research (JMLR)

Research Experience

- **Berkeley Artificial Intelligence Research (BAIR) in UC Berkeley** — Berkeley, CA, US
  Undergraduate research assistant advised by Prof. Yi Ma — *Nov 2022 - Jun 2024*
  - In general, seeking low-dimensional structures of high-dimensional natural signals, i.e. images and languages
  - Empirically investigating the properties of white-box transformers
  - Exploring mathematically interpretable deep learning architectures driven by optimizing sparse rate reduction
  - Pushing the limits of image clustering in the age of pre-trained models

Selected Awards

- **Outstanding Graduate** Shanghai, China
  Affiliation: ShanghaiTech University *Jul, 2024*
- **Merit Student** Shanghai, China
  Affiliation: ShanghaiTech University *Dec 2023*
- **Provincial First Prize for 35th National Physics Olympics Competition** Nanjing, Jiangsu, China
  Affiliation: Suzhou High School of Jiangsu Province *Sep 2018*

Invited Talks

- Presented on White-Box Transformers @ SVIP Lab, Sep 25th, 2023
- Presented on Emergence of Segmentation in White-Box Transformers @ CPAL 2024, HKU, January, 2024